

FUNCTIONS

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1)

PROGRAM:

```
mysql>
mysql> DELIMITER ;
mysql> use ten;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION customer_level(credit_limit DECIMAL(10,2))
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
->     DECLARE customer_level VARCHAR(20);
->
->     IF credit_limit > 50000 THEN
->         SET customer_level = 'PLATINUM';
->     ELSEIF credit_limit >= 10000 AND credit_limit <= 50000 THEN
->         SET customer_level = 'GOLD';
->     ELSE
->         SET customer_level = 'SILVER';
->     END IF;
->
->     RETURN customer_level;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
```

OUTPUT:

```
mysql> SELECT customer_level(60000);
+-----+
| customer_level(60000) |
+-----+
| PLATINUM               |
+-----+
1 row in set (0.00 sec)

mysql> SELECT customer_level(25000);
+-----+
| customer_level(25000) |
+-----+
| GOLD                  |
+-----+
1 row in set (0.00 sec)

mysql> SELECT customer_level(5000);
+-----+
| customer_level(5000)  |
+-----+
| SILVER                |
+-----+
1 row in set (0.00 sec)

mysql>
```

2)

PROGRAM:

MySQL 8.0 Command Line Client

```
mysql> SET @@GLOBAL.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.01 sec)

mysql> SET @@session.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE factorial(
->     IN n INT,
->     OUT result BIGINT
-> )
-> BEGIN
->     DECLARE temp BIGINT;
->
->     IF n <= 1 THEN
->         SET result = 1;
->     ELSE
->         CALL factorial(n - 1, temp);
->         SET result = n * temp;
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

OUTPUT:

```
mysql> CALL factorial(5, @result);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @result AS Factorial;
+-----+
| Factorial |
+-----+
|        120 |
+-----+
1 row in set (0.00 sec)

mysql> CALL factorial(6, @result);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @result AS Factorial;
+-----+
| Factorial |
+-----+
|        720 |
+-----+
1 row in set (0.00 sec)
```